

TEAM

CHAOS

AUTONOMOUS AI LOGISTICS
CONTROL TOWER

Cold-chain resilience • disruption response • fleet intelligence

MISSION

Protect critical cargo by turning live disruptions
into explainable autonomous recovery decisions.

TEAM CHAOS • Turn disruption into a decision.

MISSION CONTROL CREW

Suggested responsibility split

01

DIVYARAJ PUWAR

LEADER

AI Decision Engine • Architecture •
Integration • Pitch

02

BHAkti HOTWANI

COLD-CHAIN

IoT telemetry • Temperature anomaly
logic • Safety rules

03

ANJALI PATEL

FRONTEND

Control tower UI • Data visualization •
Demo experience

04

DHAIRYA JANI

BACKEND + OPT.

FastAPI • Routing logic • Fleet +
scenario simulation

DISRUPTION

AI REROUTE

COLD CHAIN

FLEET AI

When the network breaks, cargo risk compounds.

The disruption is bigger than traffic: time, cost, capacity and cold-chain exposure move together.

12–48h

POTENTIAL DISRUPTION WINDOW

2 / 4

SHIPMENTS AT RISK IN DEMO

2

IDLE REFRIGERATED ASSETS

2–8°C

CRITICAL BIOLOGICAL RANGE

THE OLD RESPONSE

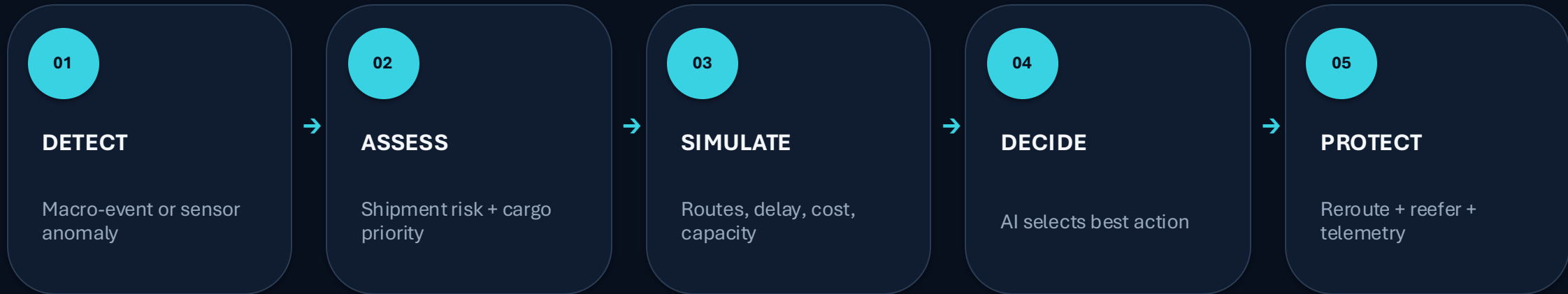
- Disruptions are discovered after they happen.
- Routing optimizes roads, not cargo condition.
- Fleet capacity can sit idle.
- Cold-chain breaches may be detected too late.

OUR RESPONSE

- Detect → prioritize → simulate → recommend.
- Optimize delay, cost, cargo and capacity together.
- Match reefers to distressed shipments.
- Monitor temperature and escalate early.

One control tower. One decision loop.

Scattered operational signals become one explainable recovery plan.

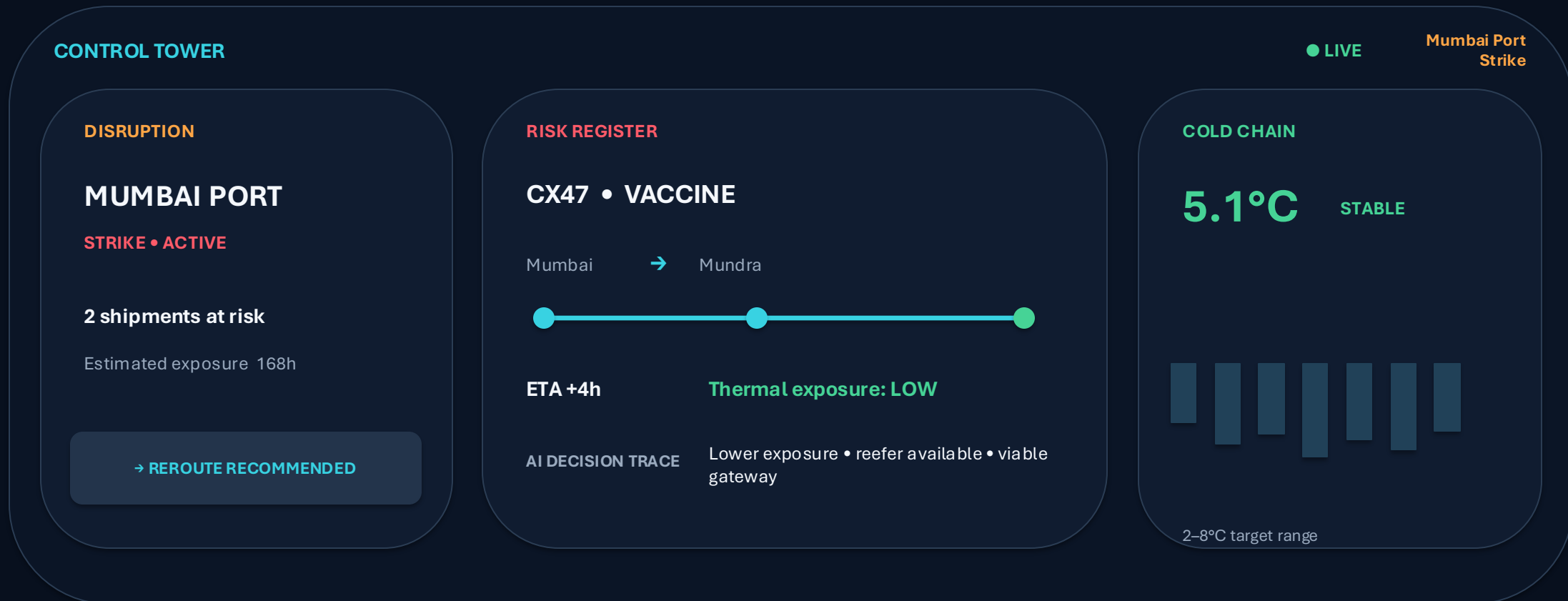


CORE PRINCIPLE

Don't just show the problem. → Calculate the best response.

The control tower sees the network as a system.

One screen connects disruption intelligence, shipment risk, fleet state and cold-chain telemetry.



From a port strike to an actionable recovery plan.

The AI evaluates alternatives instead of leaving the operator with a red alert.

AI DECISION TRACE

EVENT	Mumbai Port Strike
IMPACT	CX47 vaccine at risk
OPTION	Evaluate alternate gateways
ASSET	TRK-115 refrigerated truck
ACTION	Reroute via Mundra

WHY THIS ACTION?

LOWER EXPOSURE

- ✓ Alternate gateway is viable
- ✓ Reefer capacity is available
- ✓ Thermal window remains safe

RESULT → ranked action + explanation

Cargo condition becomes part of the routing decision.

A route is only “safe” if the shipment can survive the exposure window.

TEMPERATURE

5.1°C

● STABLE



2°C

8°C

Live telemetry stream

TEMPERATURE SAFETY LOOP

01

TELEMETRY

Live temperature stream

02

THRESHOLD

2–8°C target range

03

DETECT

Identify excursion

04

PREDICT

Estimate exposure

05

RESPOND

Escalate / reroute / hub

AI does the trade-off calculation for the operator.

Time + cost + cargo priority + thermal risk + fleet availability are evaluated together.

DECISION SCORE = Delay Risk + Thermal Risk + Cost + Capacity Penalty

SHIPMENT PRIORITY

Critical medical cargo gets higher weight

ROUTE FEASIBILITY

Blocked lanes, hubs, travel time

COLD-CHAIN RISK

Temperature + exposure window

FLEET AVAILABILITY

Idle reefer capacity for redeployment

OUTPUT → ranked actions + explanation + estimated impact

Stress-test the network before the real disruption happens.

Ask “what if?” and compare recovery strategies before the disruption becomes a failure.

SCENARIO

Mumbai Port Strike

DISRUPTION DURATION

168 HOURS

Compare strategies

- 1 Stay on current lane
- 2 Reroute via alternate gateway
- 3 Deploy idle reefer

PROJECTED IMPACT

168h

STRESS HORIZON

\$84K

EXAMPLE EXPOSURE

1

CRITICAL ACTION

The simulator turns “what if?” into measurable operational decisions.

USE CASES • capacity planning • disruption drills • cost-risk analysis

Navigation finds a road. We find a survivable plan.

Our value is the decision layer around the route.

CONSUMER NAVIGATION

- Optimizes travel between points
- Traffic + ETA are primary signals
- Limited cargo priority awareness
- No native cold-chain reasoning
- Does not allocate fleet capacity

OUR CONTROL TOWER

- ✓ Optimizes route + delay + cost
- ✓ Understands shipment criticality
- ✓ Uses temperature telemetry
- ✓ Matches idle reefers to urgent cargo
- ✓ Simulates future disruption impact

ROUTE OPTIMIZATION ≠ LOGISTICS DECISION INTELLIGENCE

A practical AI + optimization pipeline behind the command deck.

Modular layers can evolve independently as real data and telematics are added.



DEMO STACK Python • FastAPI • Optimization / Rules • IoT telemetry • Web dashboard

What makes Team Chaos different?

We combine several operational decisions into one autonomous response loop.

01

MULTI-CONSTRAINT AI

Route, delay, cost, cargo priority, thermal risk and fleet capacity.

02

COLD-CHAIN-AWARE ROUTING

A route can be rejected if cargo may not survive the exposure window.

03

DYNAMIC REEFER MATCHING

Idle refrigerated assets become emergency capacity.

04

DISRUPTION DIGITAL TWIN

Scenario testing estimates longer or cascading shocks.

05

EXPLAINABLE ACTIONS

Every recommendation shows the trade-offs behind the decision.

06

PROACTIVE DEFENSE

Act before delivery failure, not merely report failure afterward.

From a working demo to a real resilience platform.

The prototype proves the decision loop; the next step is connecting it to real operational data.

TODAY • PROTOTYPE

- ✓ Live command-deck dashboard
- ✓ Synthetic shipment + fleet data
- ✓ Disruption + reroute simulation
- ✓ Cold-chain telemetry visualization
- ✓ AI decision trace / recommendation

DEMO-READY

NEXT • SCALE

- Connect real edge telematics / GPS
- Add live disruption feeds
- Learn risk weights from historical data
- Integrate carrier + cold-hub availability
- Add human approval + audit trail

PRODUCTION PATH

TEAM CHAOS

THANK YOU

PROTECT THE CARGO.

OPTIMIZE THE NETWORK. • ACT BEFORE FAILURE.

QUESTIONS • LET'S DEMO

BoB AI Innovation Hackathon